



Hierarchical Bayes small area estimation for county-level health prevalence to having a personal doctor

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Abstract

The complexity of survey data and the availability of data from auxiliary sources motivate researchers to explore estimation methods that extend beyond traditional survey-based estimation. The U.S. Centers for Disease Control and Prevention's Behavioral Risk Factor Surveillance System (BRFSS) collects a wide range of health information, including whether respondents have a personal doctor. While the BRFSS focuses on state-level estimation, there is demand for county-level estimation of health indicators using BRFSS data. A hierarchical Bayes small area estimation model is developed to combine county-level BRFSS survey data with county-level data from auxiliary sources, while accounting for various sources of error and nested geographical levels. To mitigate extreme proportions and unstable survey variances, a transformation is applied to the survey data. Model-based county-level predictions are constructed for prevalence of having a personal doctor for all the counties in the U.S., including those where BRFSS survey data were not available. An evaluation study using only the counties with large BRFSS sample sizes to fit the model versus using all the counties with BRFSS data to fit the model is also presented.

Keywords Behavioral Risk Factor Surveillance System · Disaggregation · Hierarchical Bayes · Multiple data sources · Nested levels

1 Introduction

The U.S. Centers for Disease Control and Prevention's (CDC's) Behavioral Risk Factor Surveillance System (BRFSS) collects a wide range of health information, including whether respondents have a personal doctor. BRFSS data are used by state health departments to plan interventions, allocate scarce resources, and provide information to the general public. While the BRFSS focuses on state-level

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estimation, there is demand for county-level estimation of health indicators using BRFSS data. The ability to make the most productive use of the BRFSS data and to improve county-level planning based on BRFSS data is essential.

The overall large sample size of BRFSS allows for county-level survey-based estimation of health indicators. However, survey-based estimates can only be produced for counties with sample survey data available and are subject to high uncertainty for counties with small sample sizes. The sparsity of the survey data in some counties and the availability of data from auxiliary sources motivate researchers to explore estimation methods that extend beyond the traditional survey-based estimation. For example, model-based small area estimation (SAE) provides a principled way to combine survey and auxiliary data, while exploring the relationship between the outcome of interest and auxiliary information, and accounting for sources of uncertainty in the model components.

There are two main classes of SAE models: unit-level models, introduced in Battese et al. (1988), and area-level models, introduced in Fay and Herriot (1979). The first class of SAE models take as input unit-level survey and auxiliary data, while the second class of SAE models take as input area-level survey estimates and auxiliary data. The area-level SAE models are often preferred because of their practical applicability, while overcoming challenges related to the availability of auxiliary data, linking between the survey unit and the auxiliary source unit, inclusion of survey design effects, and others (for example, definition of unit and computational resources). These models are fit to the set of areas with sample data, but estimates can be constructed for all the areas of interest.

In this paper, an area-level SAE model is developed to combine county-level BRFSS survey data with county-level data available from auxiliary sources, for the purpose of estimating the proportions of individuals who do not have a personal care provider or doctor. The areas of interest comprise of the set of U.S. counties (3142 counties), but the model is fit only to the set of counties for which BRFSS data on having or not a personal care provider or doctor are available (3115 counties). For evaluation purposes, the model is also fit to the set of counties with BRFSS sample size of at least 500, also known as the Selected Metropolitan/Micropolitan Area Risk Trends (SMART) counties (213 counties). Model-based county-level estimates are produced for all the U.S. counties.

SAE models have been considered in the past for estimating BRFSS quantities. We first review some of the unit-level models studies. For county-level estimation of prevalence of self-reported diagnosed diabetes, Cadwell et al. (2010) developed a census region-specific model taking as input BRFSS survey data at a level defined by the cross-tabulation of age, sex, race/ethnicity, and county; this model fits under the unit-level modeling framework. For county-level estimation of chronic obstructive pulmonary disease prevalence, Zhang et al. (2014) developed a model taking as input BRFSS survey data at a level defined by the cross-tabulation of age, sex, race/ethnicity, county, and state, with extensions to finer geographies including census blocks and tracts; this model fits under the unit-level modeling framework. The model in Zhang et al. (2014) was then applied to BRFSS data on health status and access indicators, after removing sex and state from the definition of the domain level [see (Pierannunzi et al. 2016)], to BRFSS data on colorectal cancer

screening prevalence [see (Berkowitz et al. 2018)], and to BRFSS data on mammography screening rates, after removing sex from the definition of the domain level [see (Berkowitz et al. 2019)]. Holt et al. (2019) developed a similar model to the one in Zhang et al. (2014) for estimating the number of at-risk community-dwelling adults with a chronic condition (chronic obstructive pulmonary disease), while combining BRFSS and National Hurricane Center (NHC) data. As documented in the studies cited above, the current methods are not capturing the BRFSS complex survey design effects, hence are lacking at accounting for a key source of uncertainty in the survey data. Moreover, the pool of auxiliary data is constrained to BRFSS demographic variables for most of these studies. Unlike these studies, we explore a large pool of auxiliary data and adopt an area-level modeling approach that starts with BRFSS survey estimates and accounts for the sampling variability in the survey data.

Raghunathan et al. (2007) developed a joint county-level model for BRFSS and the National Health Interview Survey (NHIS) current-smoking and mammography screening rates; this model fits under the area-level modeling framework. For estimating county-level smoking and screening for female breast cancer, cervical cancer, and colorectal cancer rates, Liu et al. (2019) extended the model in Raghunathan et al. (2007) that combines BRFSS and NHIS data. Like in Raghunathan et al. (2007), we also model the arcsine-square-root-transformed survey-based county-level estimates, in order to mitigate unstable survey variances and help support a normality assumption for the survey-based county-level estimates. As a result, the sampling variances for the transformed survey estimates are functions of the effective sample sizes. For counties with small sample sizes, the authors imputed the design effect and set a lower bound for the effective sample sizes. On the other hand, to mitigate such scenarios, as well as extreme survey-based proportions, we approximate the effective sample sizes using Kish's approximation [see (Kish 1965)]. As an alternative to the arcsine-square-root transformation, Beta regressions have been considered in model-based small area estimation studies [see, for example, Wieczorek and Hawala (2011), Fabrizi et al. (2016), Janicki (2020)]. We did not investigate this approach.

Multi-fold SAE models have been studied in the literature to further account for the nested data structure of the survey data and allow for reliable estimation at various levels of aggregation. Area-level multi-fold SAE models were initially introduced in an unpublished manuscript by Fuller and Goyeneche (1998), as two-fold models, and then considered in Torabi and Rao (2014). Both of these studies employed frequentist estimation of model parameters. Erciulescu et al. (2020) specified a twofold SAE model as a hierarchical Bayes model and derived closed-form expressions for model predictions for counties with survey data. Krenzke et al. (2020) applied three-fold SAE models specified as hierarchical Bayes models to predict U.S. adult competency at the county-level. We consider Bayesian inference to be a straightforward inferential approach for multi-fold SAE models, while allowing for the construction of full posterior distributions for quantities of interest. In addition, we model county-level BRFSS and auxiliary data, while accounting for the nested structure of counties within states and states within census divisions and allowing for reliable estimation at these three levels: county, state, and census division. In this

aspect, the hierarchical Bayes SAE model developed in this paper is similar in structure to the ones considered in Krenzke et al. (2020).

The rest of the paper is organized as follows. In Sect. 2, we describe the data available for the application study and the initial survey estimation and variable selection steps necessary to prepare the model input data. The hierarchical Bayes SAE model is presented in Sect. 3, along with a framework for model fit, internal validation, estimation and prediction, and comparison. Final selected results are provided in Sect. 4 and include external validation checks. A general discussion is given in Sect. 5.

2 Data for the application study

The survey data for the application consist of micro-level BRFSS data for the reference year 2018, subject to unit-level adjustments described next. Using the BRFSS information on whether or not an individual has a personal care provider or doctor, an indicator variable is constructed having these two categories. The item nonresponse rate for this indicator variable is 0.64%. Missing values are imputed using a hot-deck imputation approach. The survey weights are adjusted using a raking procedure to align the population totals estimated from the survey to corresponding fixed controls available from the 2014–2018 American Community Survey (ACS) Summary File (also known as Detailed Tables). For counties with sample size 500 or more, the raking dimensions are age, sex, and race/ethnicity, within county. For counties with sample size less than 500, the raking dimensions are age, sex, race/ethnicity, and county, within state. Counties without sample data are excluded from raking. The resulting micro-level data are used to produce county-level survey estimates for proportions of individuals who do not have a personal care provider or doctor. Specifically, Hájek-type point estimators and associated Taylor series variance estimators are constructed for all the counties with available survey data. Let these be denoted by county-level pairs $(\hat{p}_{ijk}, \hat{V}_{ijk})$, where $i = 1, \dots, m$ indexes the census divisions, $j = 1, \dots, m_i$ indexes the states in census division i , and $k = 1, \dots, m_{ij}$ indexes the counties in state j and census division i . Let the county-level population totals from the 2014–2018 ACS Summary File be denoted by t_{ijk} .

County-level survey estimates and associated variances on the arcsine-square-root scale serve as inputs into the models described in the next section. Let these be denoted by county-level pairs $(y_{ijk}, \sigma_{ijk}^2)$, where

$$y_{ijk} := \sin^{-1} \sqrt{\hat{p}_{ijk}},$$

and

$$\sigma_{ijk}^2 := \frac{1}{4\tilde{n}_{e,ijk}}.$$

The county-level effective sample sizes are approximated as

$$\tilde{n}_{e,ijk} \approx \frac{(\sum_{a \in k} w_{ijka}^f)^2}{\sum_{a \in k} (w_{ijka}^f)^2},$$

where w_{ijka}^f is the adjusted survey weight associated with the individual a in county k . See Gabler et al. (1999) for a model-based justification of a more general formulation.

The pool of variables related to the transformed proportion of individuals who do not have a personal care provider or doctor consists of demographic characteristics (i.e., race/ethnicity, age, sex, marital status), socioeconomic status (i.e., poverty, income, employment status, occupation), education (i.e., education, English-speaking ability), location (i.e., urbanicity), immigration status (i.e., length of stay for foreign-born people, migration), health (i.e., insurance, health professions, facilities), and other (i.e., journey to work, housing unit tenure, tax, birth rate, fertility rate, infant mortality rate, crime rate, Federal aid, energy consumption). A list of data sources and auxiliary variables is provided in the Appendix A Table 5.

As the initial step in the variable selection process, we identify the most complete and less prone to error county-level auxiliary information. For this, the initial pool of variables was inspected for completeness using simple counts of missing county-level values and it was inspected for possible error using simple review of the data sources and variable definitions, raw data collected using large surveys or administrative programs being preferred over data collected using small surveys or derived using analytic methods such as models. Next, a correlation analysis is conducted to reduce the pool of auxiliary variables to those identified to be highly correlated with the transformed proportion of individuals who do not have a personal care provider or doctor and minimally correlated with other auxiliary variables in the pool. We recognize that the results from this correlation analysis could be biased (see Lahiri and Suntornchost 2015). Finally, a small set of auxiliary variables is selected using the least absolute shrinkage and selection operator (LASSO), with 20-fold cross-validation. Two choices are considered for the LASSO shrinkage/penalty parameters, both close to the optimal value that minimizes the mean cross-validated error but resulting in different numbers of selected variables. The two corresponding models constructed with these two sets of covariates will be referred to as the full model and reduced model, respectively. The full model contains a larger number of covariates than the reduced model does. Irrespective of the model, let the matrix of covariates be denoted by x_{ijk} ; its rows correspond to counties with survey sample and its columns correspond to an intercept, followed by the selected variables. The exact number of counties with survey sample and the exact number of selected variables used to fit the models are discussed in the next section.

3 Models

The overall structure of the proposed small area model follows closely the structure of the small area model for average adult proficiency presented in Krenzke et al. (2020). First, a sampling level is specified for the survey estimates on the transformed scale:

$$\text{Sampling level: } y_{ijk} | (\theta_{ijk}, \sigma_{ijk}^2) \sim N(\theta_{ijk}, \sigma_{ijk}^2),$$

where θ_{ijk} are the county-level quantities of interest on the transformed scale, i.e., the transformed proportions of individuals who do not have a personal care provider or doctor. Then, we borrow strength from auxiliary data and across small areas, while accounting for the nested structure of the data, via a linking level:

$$\begin{aligned} \text{Linking level: } \theta_{ijk} | (\beta, c_{ijk}, s_{jk}, d_k) &= x_{ijk}\beta + c_{ijk} + s_{ij} + d_i, \\ c_{ijk} | (\sigma_c^2) &\sim N(0, \sigma_c^2), \\ s_{ij} | (\sigma_s^2) &\sim N(0, \sigma_s^2), \\ d_i | (\sigma_d^2) &\sim N(0, \sigma_d^2), \end{aligned}$$

where β is a vector of unknown regression coefficients, and c_{ijk} , s_{ij} , and d_i are the county, state, and census division latent effects, respectively, assumed to follow normal distributions with zero means and unknown variances σ_c^2 , σ_s^2 , and σ_d^2 , respectively.

To fully specify the model as a hierarchical Bayes model, we adopt independent weakly informative priors for the unknown model parameters β , σ_c , σ_s , and σ_d :

$$\begin{aligned} \text{Priors: } \beta &\sim N(0, 10^4), \text{ component-wise,} \\ (\sigma_c, \sigma_s, \sigma_d) &\sim \text{Cauchy}(0, 5), \text{ component-wise.} \end{aligned}$$

These choices of prior distributions ensure that little information about the values of these parameters is provided to the model, the data (likelihood) having the major role in shaping the posterior distributions. See Krenzke et al. (2020), Browne and Draper (2006), Gelman (2006), and Polson and Scott (2012), for some recent discussions about the choice of prior distribution for the scale parameter in a univariate hierarchical Bayes model.

3.1 Fit

Four models are developed, depending on the number of counties with survey sample used and on the number of variables selected. For evaluation purposes only, we consider two sets of counties with survey data to fit the model presented above: one set comprises of all but 27 counties in the United States and corresponds to all the counties with BRFSS survey data (3114 counties and the District of Columbia), and the other set comprises of counties with BRFSS sample size of at least 500 and corresponds to all the BRFSS SMART counties (213 counties). The variable selection steps are specific to the model, so the process presented above is conducted twice, each variable selection process corresponding to the two sets of counties used in the model fit. To summarize, we fit a pair of full and reduced models for all the counties with survey sample and another pair of full and reduced models for the SMART counties. The two pairs of full and reduced model fits are compared using the widely applicable information criterion (WAIC; Watanabe 2013); smaller values indicate better model fit. Each model is specified as an R STAN object and it is fit using Markov chain Monte Carlo (MCMC), with three chains, each chain having an

initial length of 20,000 samples, of which 5000 samples are burned in. To reduce the autocorrelation in the chains used to make inference, we thin each of the chains so that every 10th iteration is kept and the rest are discarded. The final set of $R = 4500$ samples is used for inference.

3.2 Internal validation

Internal validation checks are conducted iteratively with the model development process for each model; see Erciulescu and Opsomer (2019). The Gelman-Rubin $\hat{R}$ statistic and the MC effective sample size were constructed for all the monitored parameters (regression coefficients, variance parameters, log-likelihood) and used as mixing and converge diagnostics for the MCMC sampler. Residual diagnostics were conducted to check the normality assumption, with the two sets of residuals defined as

$$\frac{y_{ijk} - x_{ijk}\hat{\beta}}{\hat{\sigma}_c^2 + \hat{\sigma}_s^2 + \hat{\sigma}_d^2 + \sigma_{ijk}^2} \text{ and } \frac{y_{ijk} - \hat{\theta}_{ijk}}{\sigma_{ijk}^2}.$$

Posterior predictive checks were conducted to further assess the adequacy of the model fit. For this, new samples $y_{ijk}^t, t = 1, \dots, T, T \leq R$ were generated from the model using the posterior samples of the model parameters. Then, the following posterior quantities were defined to test the normality and the linearity assumptions of the model, as well as the correlation between the response variable and the first covariate:

- Deviance, $T^{-1} \sum_t (y_{ijk}^t - \theta_{ijk}^t)$;
- Unscaled Residual, $T^{-1} \sum_t (y_{ijk}^t - y_{ijk})$;
- Scaled Residual, $\frac{T^{-1} \sum_t (y_{ijk}^t - y_{ijk})}{(T-1)^{-1} \sum_t (y_{ijk}^t - T^{-1} \sum_t y_{ijk}^t)^2}$;
- Identity, y_{ijk}^t ;
- Correlation, $Cor(y_{ijk}^t, x_{1,ijk})$.

For the first three quantities, median values close to zero and minimum and maximum scaled residuals with absolute values lower than three indicate no substantial model lack of fit. For the last two quantities, two test statistics were considered as follows:

- Identity, $\sum_{i,j,k} I(y_{ijk}^t \geq y_{ijk})$;
- Correlation, $\sum_{i,j,k} I(Cor(y_{ijk}^t, x_{1,ijk}) \geq Cor(y_{ijk}, x_{1,ijk}))$.

Then, posterior predictive p -values were constructed as the proportions of summary statistics calculated with sampled generated from the posterior predictive distribution that exceeded the corresponding values based on the original sample. A p -value close to 0.5 indicates that the model provides a reasonable fit to the sample data.

The results for the final model showed that convergence and mixing of the MCMC have been achieved, the resulting R posterior samples being validated for inference usage. Overall, for the final model, there were no significant departures from the normality and homogeneous variance assumptions for standardized residuals, and no substantial indication of model lack of fit using a variety of posterior predictive checks. The tuning parameters used in the model fit are specific to each model.

3.3 Prediction

Define in-sample counties as the set of counties used to fit the model. The rest of the counties of interest comprise the not-in-sample counties. The model predictions for the in-sample counties are composites of survey estimates and model-synthetic predictions based on the hierarchical Bayes model presented above. Survey estimates with large associated variance estimates (or small approximated effective sample sizes), relative to all the variance estimates, are smoothed more than the others, towards a common trend prediction that depends on the linear relationship assumed between the survey estimates and the covariates. As a result, the model predictions for such counties deviate from the survey estimates: point estimates may be either smaller or larger, and variance estimates are smaller. At the same time, also a result of the smoothing, the model predictions for counties with small variance estimates (or large approximated effective sample sizes) remain close to the survey estimates; both the point estimates and their associated variance estimates. For example, the model predictions are closer to the corresponding survey estimates for SMART counties, compared to the rest of the counties. Using the R samples $\theta_{ijk,\zeta}$, $\zeta = 1, \dots, R$, from the posterior distribution $[\theta_{ijk} || y, x, \sigma]$, posterior summaries such as means, variances, credible intervals, are constructed for the transformed proportion of individuals without a personal doctor, for all the counties.

Compared to the prediction for in-sample counties, prediction for not-in-sample counties relies more heavily on the model structure and on the covariates available for these counties. The contribution of the survey estimates to these predictions is only present via the nested structure of the model, because there is no contribution from the survey (the county-level survey estimates are either not available for these counties, or are not used as is the case with our model fitted only to the SMART counties); see Erciulescu et al. (2020) for derived expressions for the model predictions under a two-fold small area estimation model.

For a not-in-sample county k^* , where the county belongs to a state j already represented in the in-sample set of counties, we generate R samples $\theta_{ijk^*,\zeta}$, $\zeta = 1, \dots, R$, from the distribution assumed in the linking model with parameters evaluated at the R posterior samples, $N(x_{ijk^*}\beta_\zeta + s_{ij,\zeta} + d_{i,\zeta}, \sigma_{c,\zeta}^2)$. If the not-in-sample county k^* belongs to a state j^* for which no other county component is represented in the in-sample set of counties, but it belongs to a census division i already represented in the in-sample set of counties, we generate R samples $\theta_{ij^*k^*,\zeta}$, $\zeta = 1, \dots, R$ in two steps: first, we generate state latent effect samples $s_{ij^*,\zeta}$ from $N(0, \sigma_{s,\zeta}^2)$; second, we generate R samples $\theta_{ij^*k^*,\zeta}$ from $N(x_{ij^*k^*}\beta_\zeta + s_{ij^*,\zeta} + d_{i,\zeta}, \sigma_{c,\zeta}^2)$. Finally, if the not-in-

sample county k^* belongs to a state j^* that belongs to a census division i^* for which no other county component is represented in the in-sample set of counties, we generate R samples $\theta_{i^*j^*k^*,\zeta}$, $\zeta = 1, \dots, R$ in three steps: first, we generate census division latent effect samples $d_{i^*,\zeta}$ from $N(0, \sigma_{d,\zeta}^2)$; second, we generate state latent effect samples $s_{j^*,\zeta}$ from $N(0, \sigma_{s,\zeta}^2)$; third, we generate R samples $\theta_{i^*j^*k^*,\zeta}$ from $N(x_{i^*j^*k^*}\beta_\zeta + s_{j^*,\zeta} + d_{i^*,\zeta}, \sigma_{c,\zeta}^2)$. Note that this final scenario is included here for methodology completion, but it was not encountered in the application modeling process.

To construct model predictions on the original scale, we back-transform the posterior samples using a squared-sin transformation, $\theta_{ijk,\zeta}^f := (\sin(\theta_{ijk,\zeta}))^2$, for all the in-sample and not-in-sample counties. Then, county-level posterior summaries such as means, variances, credible intervals are constructed using the R samples $\tilde{\theta}_{ijk,\zeta} \in \{\theta_{ijk,\zeta}, \theta_{ijk,\zeta}^f\}$.

State-level, census division-level, and nation-level quantities are constructed as aggregates of the county-level quantities on the original scale, with the county-level population totals t_{ijk} serving as aggregation weights. Posterior summaries for these quantities are then constructed using the corresponding R samples. The state-level samples are defined as $\theta_{ij,\zeta}^f := \sum_k t_{ijk} \theta_{ijk,\zeta}^f (\sum_k t_{ijk})^{-1}$, the census division-level samples are defined as $\theta_{i,\zeta}^f := \sum_{k,j} t_{ijk} \theta_{ijk,\zeta}^f (\sum_{k,j} t_{ijk})^{-1}$, and the nation-level samples are defined as $\theta_\zeta^f := \sum_{k,j,i} t_{ijk} \theta_{ijk,\zeta}^f (\sum_{k,j,i} t_{ijk})^{-1}$.

The predictive power of the models is tested using cross-validation. For this, the in-sample counties are divided into two groups with similar distribution of sample sizes. Each model is fitted to one of the two groups of counties, in turn, and the counties in the other group are considered not-in-sample. The model predictions for the not-in-sample counties are then compared against the survey estimates, for groups of counties with large sample size: the sum of squared differences between the survey estimates and the model predictions is constructed for each model and these values are compared across models, lower values indicating better model predictive power.

4 Results

Variable selection and cross-validation results are presented in Tables 1 and 2, for the models fitted to all the counties with sample data and for the models fitted to the SMART counties, respectively. The auxiliary variables selected in the models fitted to all the counties with sample data are available at the county-level from ACS, Small Area Health Insurance Estimates (SAHIE), or Statistics of Income (SOI). The first six variables listed in Table 2 are available at the county level, but the rest are only available at the state level. Compared to the auxiliary data selected for the models fitted to all the counties with sample data, data from three additional government agencies were selected for the models fitted to the SMART counties: the National Center for Health Statistics (NCHS), the National Highway Traffic Safety Administration (NHTSA), and the Energy Information Administration (EIA). Recall Table 5

Table 1 Variable selection and cross-validation results for the models fitted to all the counties with sample data

Variable	Source	Full model	Reduced model
Proportion of population age 25+: with high school diploma, no college	ACS	X	X
Proportion of Hispanics	ACS	X	X
Proportion of Non-Hispanic Whites	ACS	X	X
Proportion of Native Americans	ACS	X	
Proportion of owner occupied housing unit	ACS	X	X
Proportion of population 18+: in different house in the past year	ACS	X	
Proportion of population 18+: in different state in the past year	ACS	X	
Proportion of uninsured population	SAHIE	X	X
Proportion of returns with taxable Social Security benefits per person	SOI	X	
Sum of squared differences between model predictions and survey estimates			
213 counties with sample size ≥ 500		0.350	0.416
963 counties with sample size ≥ 100		4.092	4.244

The variables included in the model are indicated by X

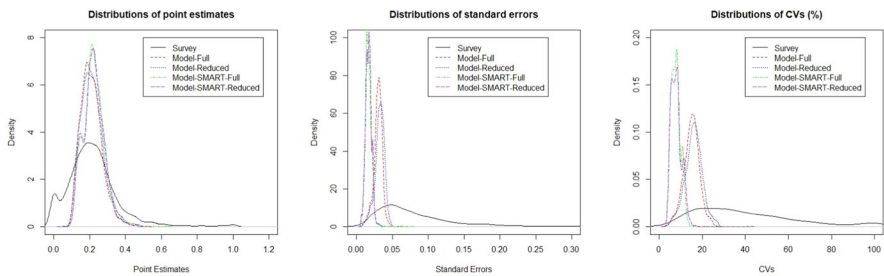
Table 2 Variable selection and cross-validation results for the models fitted to the SMART counties

Variable	Source	Full model	Reduced model
Proportion of Hispanics	ACS	X	X
Proportion of Non-Hispanic Whites	ACS	X	X
Proportion of population 55–64 years old	ACS	X	X
Proportion of population 18+: in different state in the past year	ACS	X	
Proportion of uninsured population	SAHIE	X	X
Proportion of returns with taxable Social Security benefits per person	SOI	X	
Birth rate (live birth as a proportion of total population)	NCHS	X	X
Per capita energy consumption	EIA	X	
Traffic fatalities per 100 million vehicle miles	NHTSA	X	X
Sum of squared differences between model predictions and survey estimates			
56 counties with sample size ≥ 1000		0.094	0.109
213 counties with sample size ≥ 500		0.409	0.483

The variables included in the model are indicated by X

Table 3 Model comparison results

Model	WAIC
All counties with sample data—full	− 0.7198
All counties with sample data—reduced	− 0.7177
SMART counties—full	− 2.0346
SMART counties—reduced	− 2.0117

**Fig. 1** Comparison of county-level point estimates, standard errors, and CVs; models fitted to all the counties with sample data, models fitted to the SMART counties, and survey

in Appendix A with the information on the data sources. The full models have better predictive power than the reduced models, having smaller sum of squared differences between the model predictions and the survey estimates. The results from the model comparison using WAIC are also in favor of the full models, as shown in Table 3. Despite the small differences in the WAIC values, the direction from smallest to largest is comforting, indicating that the models with better predictive power have also better fits, among the group of models considered in the comparisons.

For the counties with available survey estimates, we compare the distributions of the survey estimates and the model predictions produced using the four models under investigation. The results are illustrated in Fig. 1. There are minor differences between the full and reduced models for each of the two pairs considered. Based on these results and the results from the model comparison and cross-validation, we conclude that the reduced models are no longer of interest.

As illustrated in Fig. 1, the point estimates are comparable across the different sources considered in this comparison, with exception being the main mode of the distributions: the model predictions fall in a narrower range than the range of the survey estimates, as a result of smoothing, with the model predictions based on the models fitted to the SMART counties being smoothed towards slightly larger values than the other model predictions. The uncertainty in the model predictions based on the models fitted to the SMART counties is noticeably smaller than both the uncertainty in the survey estimates and the uncertainty in the model predictions based on the models fitted to all the counties with sample data. This result is a consequence of constructing a much larger number of not-in-sample model predictions under the approach where only the SMART counties are used in the model fit, while ignoring the survey data available for the non-SMART counties.

To further evaluate the model fits using different sets of counties, we compare the estimates for large geographies: census divisions and nation. Pairwise comparisons of the survey and model predictions for the high levels of aggregation are presented in Table 4. Note that for most of the census divisions, there is closer agreement between the survey estimates and the model predictions based on the model fitted to all the counties with sample data than the model predictions based on the model fitted to the SMART counties. One possible explanation for this result is that SMART counties, despite their relatively large sample sizes, might not always be representative for some census regions; note, for example, the discrepancy in the point estimates for East South Central. In addition, for the nation and most of the census divisions, the standard errors of the model predictions based on the model fitted to the SMART counties are larger than the survey standard errors. This result is a consequence of constructing a large number of positively highly correlated not-in-sample model predictions under the model fitted to the SMART counties only, which impacts the variance estimates at aggregated levels. Hence, we identify the model fitted to all the counties with sample data to be the preferred or final model. A map of all the county-level model predictions constructed based on the final model is illustrated in Fig. 2.

Finally, we note that the covariates selected do not present a very strong linear relationship to our outcome of interest (the proportion of individuals who do not have a personal care provider or doctor); the R squared for a multiple linear regression fitted to the county-level survey estimates and using the set of covariates selected for the model fitted to all the counties with sample data is approximately 0.08. Hence, it is important to use all the available county-level survey estimates in fitting the model, so that the number of not-in-sample predictions is as small as possible.

Table 4 Comparison of nation-level and census division-level point estimates (PEs), standard errors (SEs), and coefficients of variation (CVs); models fitted to all the counties with sample data (Model Nation), models fitted to the SMART counties (Model SMART), and survey

Nation/census division	Model nation		Model SMART		Survey	
	PE	SE	PE	SE	PE	SE
Nation	0.228	0.001	0.229	0.003	0.230	0.001
New England	0.144	0.003	0.140	0.003	0.143	0.003
Middle Atlantic	0.182	0.003	0.181	0.006	0.185	0.004
West North Central	0.224	0.003	0.222	0.005	0.228	0.003
South Atlantic	0.240	0.003	0.240	0.008	0.242	0.003
East South Central	0.221	0.004	0.241	0.014	0.229	0.004
West South Central	0.289	0.006	0.306	0.008	0.295	0.007
Mountain	0.280	0.003	0.277	0.004	0.280	0.004
Pacific	0.252	0.004	0.241	0.005	0.252	0.004

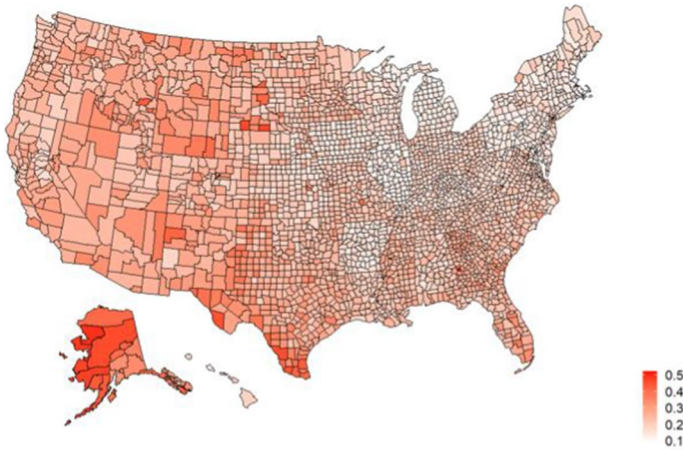


Fig. 2 Map of county-level model predictions: proportions of individuals with no personal care provider or doctor

4.1 External validation

Various external validation checks are considered for the final model. For the in-sample counties, we compare the distributions of the survey estimates and the model predictions in terms of point estimates and coefficients of variation (CVs). The results are illustrated in Fig. 3. Note that the ranges of the model predictions are narrower than the ranges of the survey estimates as a result of smoothing. The main modes of the overall distributions of point estimates overlap, with no noticeable difference between the distributions for in-sample county-level model predictions and all model predictions (for in-sample counties and not-in-sample

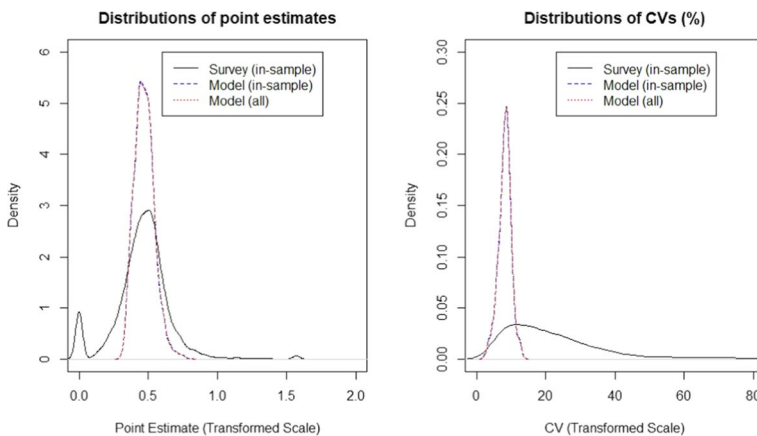


Fig. 3 Comparison of county-level model predictions and survey estimates: point estimates and CVs; transformed scale

counties). On the other hand, the main modes of the overall distributions of CVs are shifted to the left for the model predictions, compared to the survey estimates, as a result of model-based decrease in uncertainty.

Next, we investigate the distribution of the differences between the survey estimates and model predictions, and the ratios of survey to model standard errors, both relative to the county sample sizes. The results are illustrated in Fig. 4. A dotted line is included in the plots to indicate the group of SMART counties: points to the right of the line correspond to SMART counties. Note that the survey estimates and model predictions are comparable for larger counties, while for counties with smaller sample sizes, the estimates differ: model predictions deviate from the survey estimates because the latter are not sufficiently precise, and the model standard errors/CVs are smaller than the survey standard errors/CVs (on the transformed scale) as a result of model shrinkage. The shrinkage effects are observed on the scale on which the models are fitted, i.e., the transformed proportions using the arcsine-square-root transformation, and are diminished after the back transformation to the original scale, i.e. the proportions. As a consequence, the model standard errors/CVs are smaller than the survey standard errors/CVs for most, but not all of the counties used in the model fitting process.

Pairwise comparisons of the survey estimates and model predictions for the high levels of aggregation are presented in Table 4. Note that there is close agreement between the point estimates and standard errors, with slight reduction in the standard errors for some areas: census divisions Middle Atlantic, West South Central, and Mountain. The agreement in the point estimates suggests there are no necessary changes in the model specification. The reduction in the standard errors is a result of the model shrinkage, and its small magnitude is expected because at these high levels (nation and census division) the survey estimates are precise.

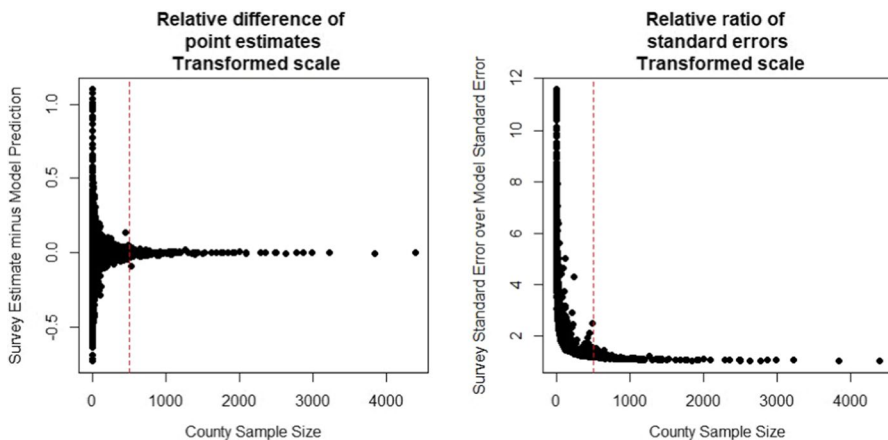


Fig. 4 Comparison of county-level model predictions and survey estimates: difference in point estimates and ratio of standard errors, relative to the sample size; transformed scale

As a final step in the external validation process, we construct the percentage of counties with non-overlapping 95% uncertainty intervals for the survey estimates and model predictions defined as

$$y_{ijk} \pm 2\sigma_{ijk}(\text{transformed scale}) \text{ and } (\tilde{\theta}_{ijk,(0.025)}, \tilde{\theta}_{ijk,(0.975)}),$$

respectively. For large counties, one would expect to see few or no non-overlapping intervals because the survey estimates in these counties are considered sufficiently precise and the model predictions should not deviate significantly from them. Since the survey estimates for smaller counties are not as reliable as the ones for larger counties, one can expect to see more non-overlapping intervals. For the SMART counties, all the uncertainty intervals overlap. Among the counties with sample sizes greater than or equal to 30, there are less than 2% non-overlapping uncertainty intervals. Among all the counties with sample, there are less than 4% non-overlapping uncertainty intervals. On the transformed scale, most (31 out of 40) of the non-overlapping uncertainty intervals correspond to counties with survey estimates of 0 or 1 on the original scale. This is expected, because the model predictions for counties with smaller sample sizes are shrunk more toward the common trend, than the large counties are, resulting in significantly different estimates from the survey estimates.

5 Discussion

In summary, we developed model-based small area estimation methodology for the proportion of individuals with no personal care provider or doctor in the 2018 population. The models combine survey data from the BRFSS with data from other sources, while accounting for the error in the survey data and the nested structure of the data. Model predictions were constructed for all the counties in the nation, irrespective of the availability of survey data. For counties where survey estimates are available, the model predictions are more precise than the survey estimates. For evaluation purposes, the models were developed using either all the counties with survey data or only the counties with sample sizes at least 500. The former was preferred because it resulted in more accurate point estimates than the latter. At higher levels of aggregation, i.e. census division and nation, there was close agreement between the model predictions and the survey estimates, indicating that there was no need for benchmarking adjustments to the model predictions. Code scripts for model specification and fit are provided as supplementary material in Appendix B.

As part of the model development, we investigated other models and compared their performance to the performance of the model presented in this manuscript. Among these alternative model specifications, we investigated area-level models for the BRFSS survey estimates on the original scale or transformed using other transformations: square root and logit. Similarly to the transformation adopted for the proposed model, the square root and the logit transformations help relax a normality assumption for the survey estimates on the original scale. However, the logit transformation is only applicable to survey estimates greater than zero and smaller than one. Moreover, for the original scale of these two alternative transformations, the

estimated variances associated with the survey estimates would not be defined for all the counties, either due to sample sizes or due to the transformation. Finally, unlike the transformation adopted for the proposed model, none of these three alternative model specifications would stabilize the sampling variances, so a smoothing step would be necessary, too.

Lacking unit-level auxiliary data for the entire population, we did not investigate any unit-level models. However, we did consider a hybrid-level model, where the sampling level is specified at the unit level and the linking level is specified at the area (county) level. As an initial step, we ignored the survey design effect and the imputation. That is, we specified the sampling level for the BRFSS unit-level raw data. The computational challenges led us to quickly set aside a nation-wide model, and instead fit a model to all the data available for one state. As part of the internal validation process, we noticed residuals with heavy-tailed distribution, so the model specification indicated there was room for improvement. Also, the state-level (aggregated) model prediction was significantly smaller than the state-level survey estimate, again indicating the model specification can be improved.

Future investigations may include model extensions and alternative specifications, and closer attention to hybrid modeling: (1) Beta regressions with extensions for 0 and 1 estimates for proportions; (2) Bayesian Lasso for conducting variable selection and parameter estimation simultaneously; (3) multivariate specifications, for example for modeling the BRFSS proportion of individuals who do not have a personal care provider or doctor and the proportion of individuals without health insurance jointly; and (4) measurement error specifications, to account for the uncertainty in the county-level covariates.

Appendix A Auxiliary data pool

See Table 5.

Table 5 Auxiliary data pool

Source	Auxiliary variable
Census Bureau American Community Survey (ACS)	5-year estimates (2013–2017) of socioeconomic demographic, and housing characteristics
Census Bureau Small Area Income and Poverty Estimates (SAIPE)	Small area estimates of selected income and poverty statistics
Census Bureau Small Area Health Insurance Estimates (SAHIE)	Health insurance coverage status by selected economic and demographic characteristics
U.S. Department of Agriculture (USDA) Economic Research Service	Classification of counties into metro and non-metro (OMB subdivided counties into three metro and six non-metro categories)
Bureau of Labor Statistics (BLS) The Local Area Unemployment Statistics (LAUS) program	Monthly and annual employment, unemployment, and labor force data
Bureau of Economic Analysis (BEA) Centers for Disease Control and Prevention Division of Diabetes Translation (DDT)	Estimates of personal income for local areas updated statistics about diabetes
Centers for Medicare & Medicaid Services (CMS) Geographic variation public use file	Utilization and quality of health care services for the Medicare fee-for-service population
The Internal Revenue Service The Statistics of Income (SOI) Data	Income and tax data such as number of tax returns, returns with unemployment compensation, returns with taxable social Security benefit, adjusted gross personal income, personal unemployment compensation amount, and personal taxable social Security benefit amount, etc.
Health Resources and Services Administration Area Health Resources Files (AHRF)	Data on health care professions, health facilities, population characteristics, economics, health professions training, hospital utilization, hospital expenditures, and environment
National Center for Health Statistics (NCHS)	Birth rate and infant mortality rate
Federal Bureau of Investigation (FBI)	Crime rates
National Highway Traffic Safety Administration (NHTSA)	Traffic fatalities
U.S. Energy Information Administration (EIA)	Energy consumption height

Appendix B STAN code

Model specification

```
<< save the model specification as a .stan object; for example, call it HM3folduniv.stan >>
data {
  int<lower=0> mc;      // number of counties
  int<lower=0> ms;      // number of states
  int<lower=0> md;      // number of census divisions
  real y[mc];          // county-level survey estimates (transformed scale)
  real<lower=0> se[mc]; // county-level survey standard errors (transformed scale)

  int<lower=0> p;       // number of regression coefficients; linking level
  real x[mc,p];        // model matrix
  int sind[mc];        // state indicator
  int dind[mc];        // census division indicator
}
parameters {
  real beta[p]; // regression coefficients
  real c[mc];   // county-level latent effects
  real s[ms];   // state-level latent effects
  real d[md];   // census division-level latent effects
  real<lower=0> sigmac; // county-level latent effects standard deviation
  real<lower=0> sigmas; // state-level latent effects standard deviation
  real<lower=0> sigmad; // census division-level latent effects standard deviation
}
transformed parameters {
  real theta[mc]; // county-level quantities of interest
  for (i in 1:mc) { // loop over all the counties
    theta[i] = dot_product(beta, x[i,]) + c[i] + s[sind[i]] + d[dind[i]]; // linking level
  }
}
model {
  // prior distributions
  for (i in 1:p) // loop over all the regression coefficients
    beta[i] ~ normal(0, 10^2) // assign prior distribution to each regression coefficient

    sigmac ~ cauchy(0,5); // assign prior distribution to the county-level latent effects standard deviation
    sigmas ~ cauchy(0,5); // assign prior distribution to the state-level latent effects standard deviation
    sigmad ~ cauchy(0,5); // assign prior distribution to the census division-level latent effects standard deviation

    c ~ normal(0, sigmac); // assign prior distribution to the county-level latent effects
    s ~ normal(0, sigmas); // assign prior distribution to the state-level latent effects
    d ~ normal(0, sigmad); // assign prior distribution to the census division-level latent effects

    for (i in 1:mc) { // loop over all the counties
      y[i] ~ normal(theta[i], se[i]); // sampling level
    }
}
generated quantities{
  vector[mc] log_lik; // log likelihood
  for (i in 1:mc) log_lik[i] = normal_lpdf(y[i] | theta[i], se[i]); // compute log likelihood
}
```

Model fit

```
### model input: survey and auxiliary data, for in-sample counties
data <- list(mc=mc, ms=ms, md=md, sind=sind, dind=dind, y=phatt1a, se=phatset1a, x=Xs, p=p)
### model fit
# install the necessary R packages and libraries, and assign computation to all available cores
require(rstan)
options(mc.cores = parallel::detectCores())
rstan_options(auto_write = TRUE)
require(coda)
require(MASS)
# run the STAN model and save it as an R object; for example, call it brfssmodel3folduniv.stan
system.time( ## print the running time
  brfssmodel3folduniv.stan <- stan('HM3folduniv.stan', ## call the STAN model
    data=data, ## call the data
    chains=3, ## use three MC chains
    iter=20000, ## use 20000 iterations for each chain, as a starting point
    warmup=5000, ## drop the first 5000 iterations from each chain
    thin=10, ## keep only the tenth iteration in each chain
    seed=20201123, ## set the random number generator seed
    control=list(max_treedepth=15))) ## tune the sampling parameter
```

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